MIL-A-83376A(USAF)
28 June 1978
SUPERSEDING
MIL-A-83376 (USAF)
26 July 1972

MILITARY SPECIFICATION

ADHESIVE BONDED METAL FACED SANDWICH STRUCTURES, ACCEPTANCE CRITERIA

This specification is approved for use by the Air Force Materials Laboratory, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1 1 Scope This specification establishes the acceptance criteria and inspection requirements for adhesive bonded sandwich structures including the metal-to-metal bonding found in these structures (See 6 1)

1 2 Classification The classification of adhesive bonded structures specified in this specification shall be of the following types

Type I - Components which are fracture or fatigue critical as defined in MIL-I-6870 components, the single failure of which would cause significant danger to operating personnel or would result in an operational penalty This includes loss of major components, loss of control, unintentional release, inability to release armament stores, or failure of weapon installation components

Type II - All components not classified as Type I

1 2.1 Structural classification The structural classification, Type I or Type II shall be specified on the engineering drawing. All structures shall be considered Type I unless otherwise specified on the drawing.

1 2 2 Structural defects Structural defects specified in this specification shall be face-to-core voids and disbonds, metal-to-metal voids, bond voids, porosity, core splice voids, core splice gaps, dents and markoffs These defects are defined in Section 6 of this specification

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Air Force Materials Laboratory, MXA, WPAFB, Ohio 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

FSC 8040

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:51Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-6870	Inspection Program Requirements, Nondestructive Testing, For Aircraft and Missile Materials and Parts.
MIL-A-83377	Adhesive Bonding (Structural) for Aerospace Systems, Requirements

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

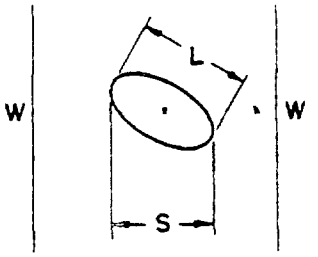
3.1 General. The contractor shall specify the allowable defects on the engineering drawing or in the process specification. If this is not so specified by the contractor, the requirements stated in Section 3 shall apply.

3.1.1 Adhesive bonded metal faced structure shall be inspected to the requirements as stated in 3.1.

3.2 Defects in Metal-to-Metal bonds in metal faced sandwich assemblies. The contractor shall specify the allowable defects on the engineering drawing or in the manufacturing process specification. If this is not so specified by the contractor the requirements stated in Section 3, herein shall apply.

3.2.1 Defect dimensions. When measured as shown in Figure 1, individual disbonds or voids shall not have any dimension greater than 15 times the thickness of the thinnest adherend, or 1 0 inch whichever is smaller or a "S" dimension greater than 15 percent of the shortest "W" through the defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:51Z
Check the source to verify that this is the current version before use.



S is the width of void in the "W" direction.
L is the longest dimension of void.
W is the smallest dimension of the metal-to-metal bond area containing the void.
The width of void is in the "W" direction.
When "W" is less than 2 inches, "S" shall be less than 0.25 inch.

Figure 1. Dimensions.

3 2 2 Permissible areas.

3 2 2.1 The maximum permissible area of an individual void or disbond is as follows

(a) Type I structure 0.375 square inches, and
(b) Type II structure 0 500 square inches.

3 2 2 2 The minimum permissible spacing distance as measured in Figure 2 between void boundaries is as follows:

(a) Type I structure 4 times the largest void or disbond dimension measured on a line between the centers of the two voids
(b) Type II structure minimum distance is 3 times the void or disbond dimension.

3 2 2 3 No voids, disbonds, or porosity shall be permitted within 0 125 inch of any bond joint edge.

3 2.2.4 The aggregate unbonded length of disbonds along any straight area, between two parallel lines 1/4 inch apart shall not exceed 15 percent of the length of the bond line, when measured along the direction of the parallel lines.

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:51Z
Check the source to verify that this is the current version before use.

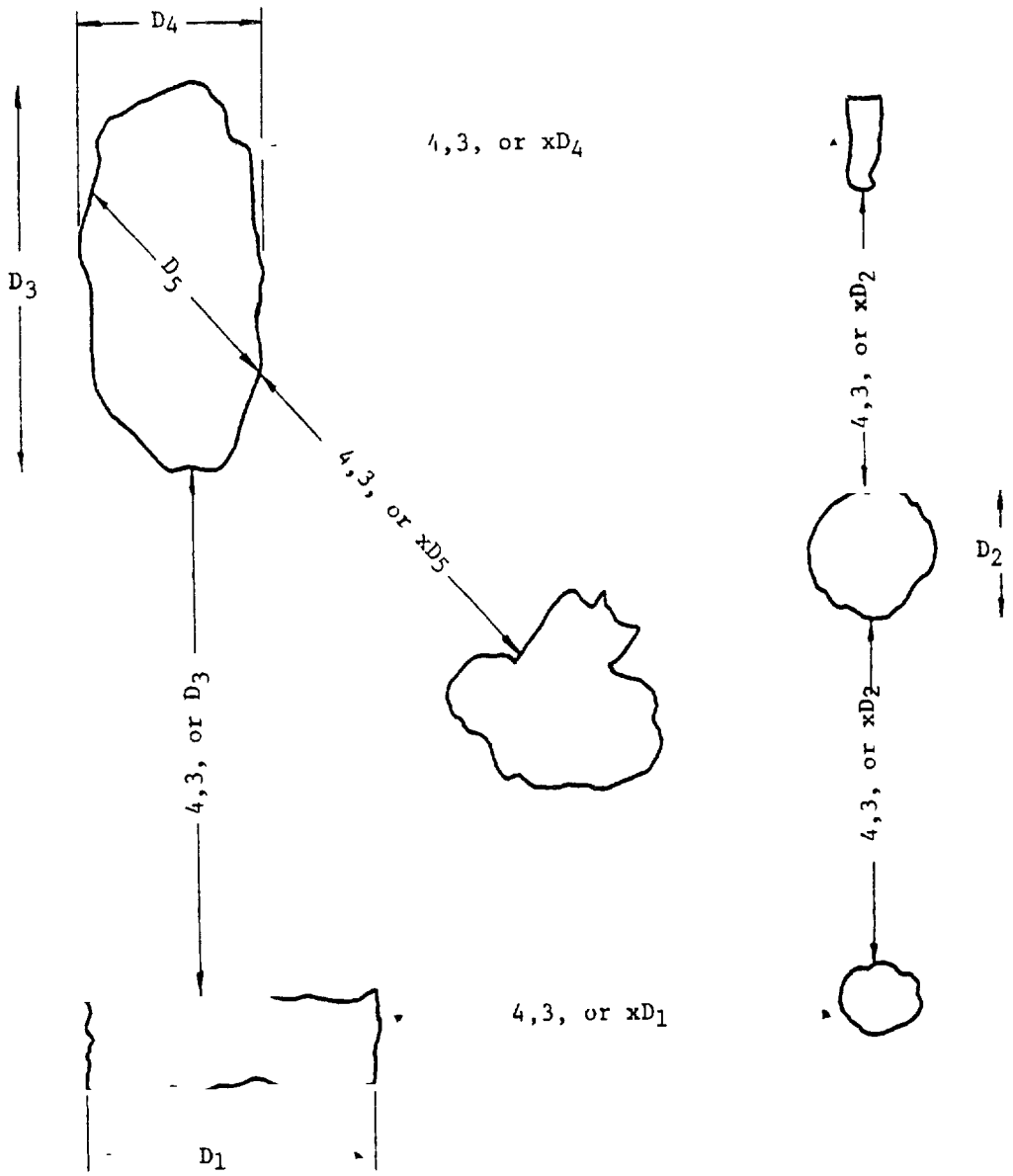


Figure 2. Measuring disbond spacing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:51Z
Check the source to verify that this is the current version before use.

3.2.2.5 The maximum permissible void or disbond area of the metal-to-metal bond area within an assembly is 1 percent on Type I structure, and 2 percent on Type II structure

3.2.2.6 The maximum permissible disbond area of the metal-to-metal bond area contained within a 6 inch diameter circle is 8 percent of that area so contained for Types I and II structures.

3.3 Face-to-core bonds. Voids or disbonds shall be limited to 0.5 inch dimaeter for Type I and 0.75 inch diameter for Type II face-to-core or doubler-to-core bonds.

3.4 Core splice, core-to-edgemember, or core-to-insert bonds.

3.4.1 The maximum permissible core gap that is filled with adhesive or the equivalent shall be equal to 0.125 inch for Type I structure and one cell diameter of the largest cell common to the core joint or 0.25 inch, whichever is smaller for Type II structure, as shown in Figures 3, 4, and 5

3.4.2 The maximum permissible core joint void or disbond dimension shall be 0.5 inch or 3 adjacent cells, whichever is smaller, per linear foot and shall not exceed 5 percent of core to core bond area (splice height times splice length)

3.4.3 The maximum core to beveled edge member voids, disbonds and gaps shall be as shown in figure 6.

3.5 Sandwich face dents

3.5.1 Dents Dents shall not exhibit creases, sharp edges, or wrinkles as shown in Figure 7.

3.5.1.2 The dent area shall be inspected for disbonds in accordance with MIL-A-83377.

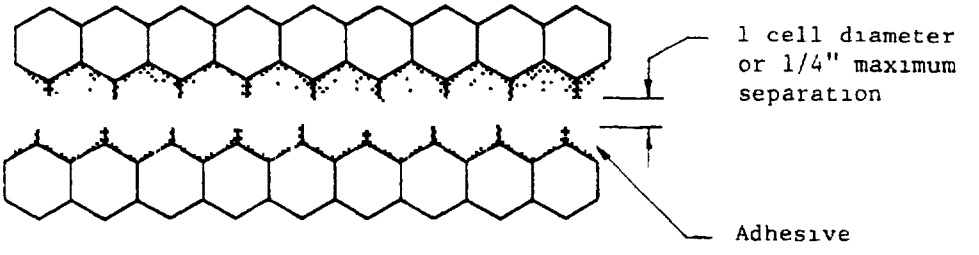


Figure 3. Core splice (Type II structure)

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:51Z
Check the source to verify that this is the current version before use.

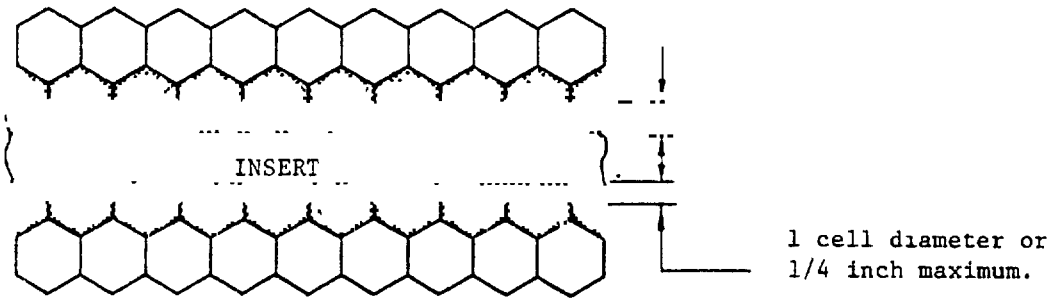


Figure 4. Core to insert (Type II structure).

NOTE: 1. The maximum void or disbond 1/2 inch or 3 adjacent cells, whichever is smaller, per linear foot.

2. Total void or disbond must not exceet 5 percent of total core to insert bond area.

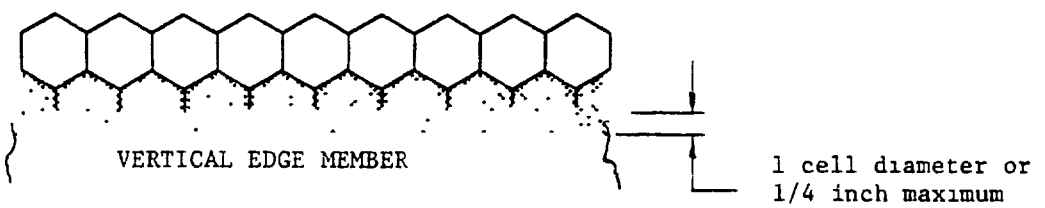


Figure 5. Core to vertical edge member (Type II Structure).

NOTE: 1. Adhesive maximum void or disbond 1/2 inch or 3 adjacent cells, whichever is smaller, per linear foot.

2. Total disbond must not exceed 5 percent of total core to edge member bond area.

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:51Z
Check the source to verify that this is the currentversion before use.

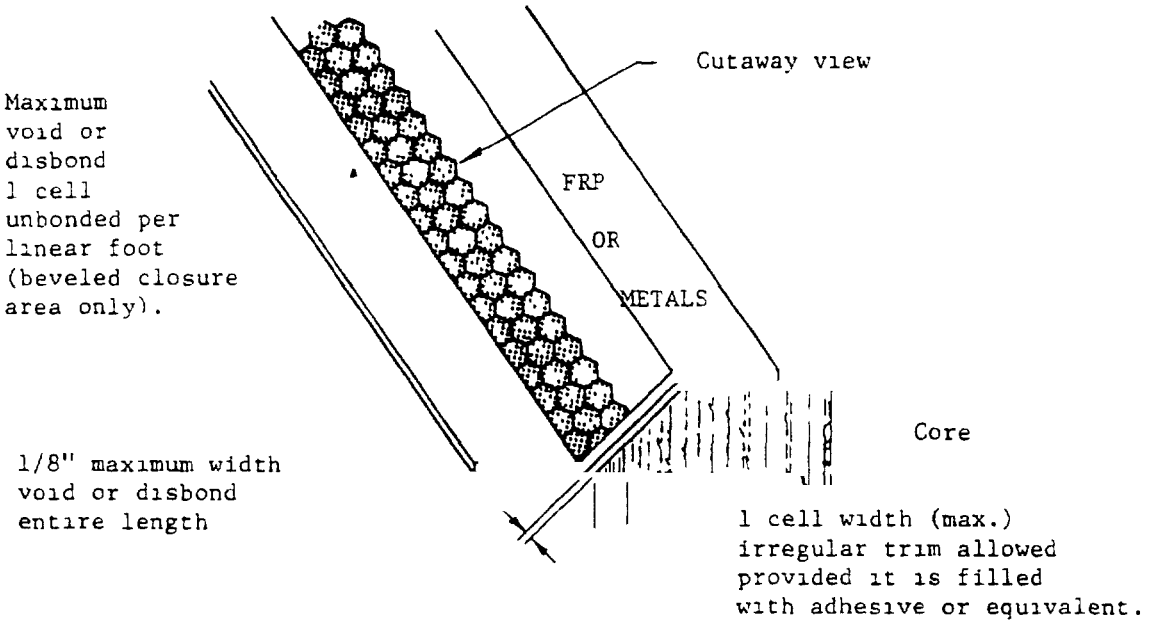


Figure 6. Core to beveled edge member.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:51Z
Check the source to verify that this is the current version before use.

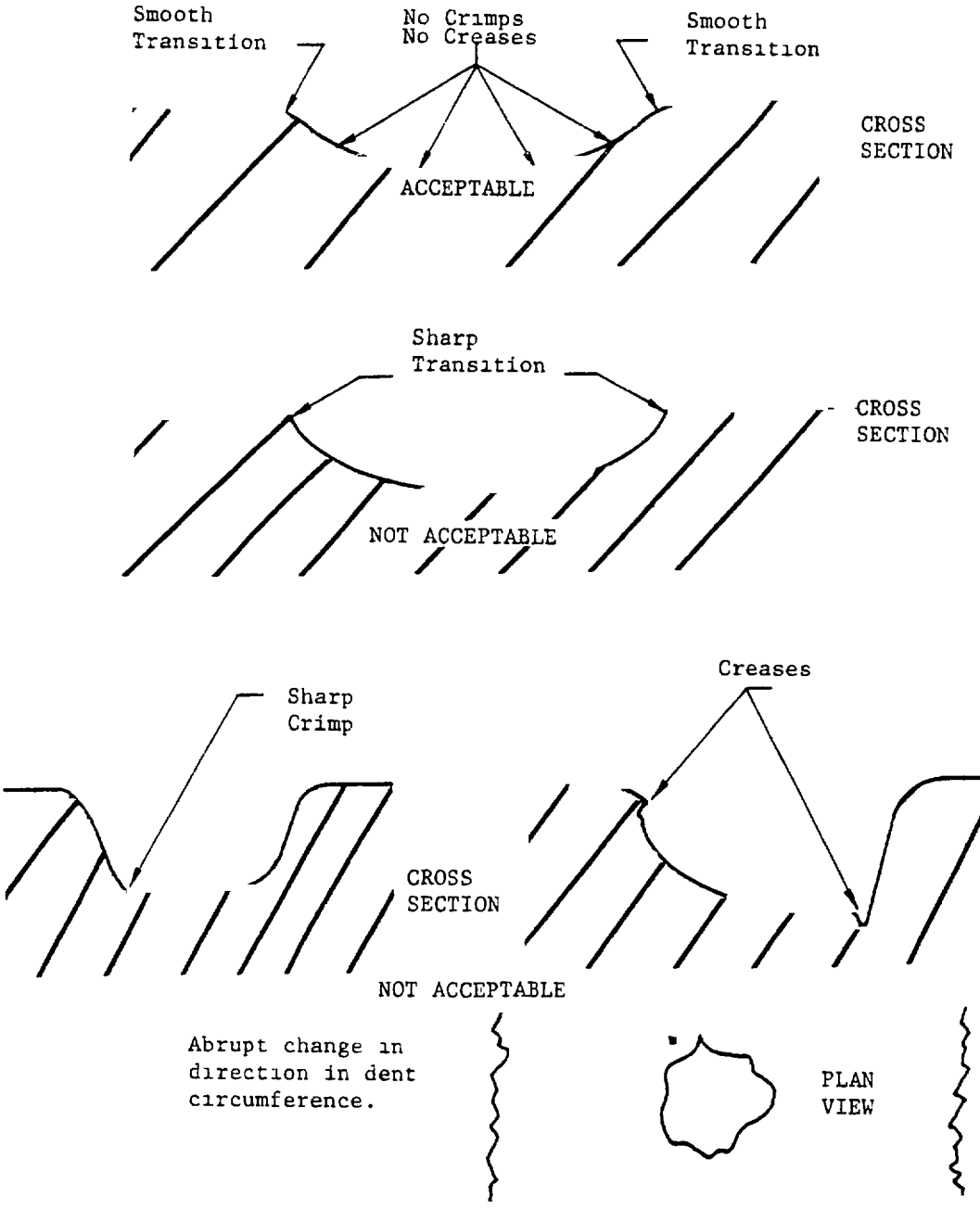


Figure 7. Sandwich face dents.

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:51Z
Check the source to verify that this is the current version before use.

3.5.2 Dimensions.

3.5.2.1 Diameter. The maximum permissible dent diameter is 1.5 inches for Type I structure and 2 inches for Type II structure.

3.5.2.2 Depth. The maximum permissible dent depth is 0.006 inches for Type I structure and 0.020 inches for Type II structure.

3.5.2.3 Areas. The maximum permissible dent area within an assembly is 1 percent on Type I structure and 2 percent on Type II structure.

3.5.2.4 Spacing. The minimum permissible distance between dent centers is 6.0 inches for Type I structures and 4.0 inches for Type II structure

3.5.2.4.1 The minimum permissible distance between dent center and the edge of solid members is 10.0 inches for Type I structure, and 6 inches for Type II structure.

3.5.3 Dent limitations. Dent limitations as required for cosmetic purposes shall be established by the contractor's quality assurance.

3.6 Face markoff. Markoff requirements shall be in accordance with Figure 8

3.7 Nondestructive inspection.

3.7.1 Assemblies shall be nondestructive inspected for adherence to the requirements of this specification and in accordance with MIL-A-83377.

3.7.2 Nondestructive inspection physical standards shall be constructed to incorporate acceptance criteria. Standards shall be of similar construction as the parts to be inspected, and shall contain voids or disbonds of the maximum allowed sizes. Nondestructive process procedures shall be approved as specified in MIL-A-83377.

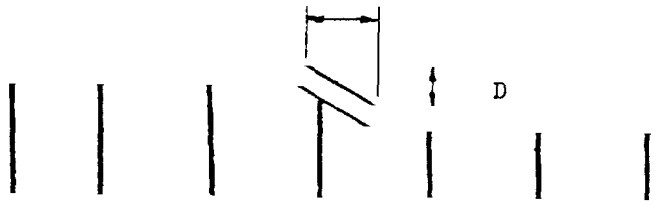
3.8 Destructive inspection.

3.8.1 Destructive inspection of an assembly shall be conducted as specified in MIL-A-83377 to ensure that the bonded components meet the criteria contained in this specification.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:51Z
Check the source to verify that this is the current version before use.



$\frac{D}{L}$ 0.006 inch/inch

Maximum markoff allowed Type I. 0.006 inch/inch

Figure 8. Markoff.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:51Z
Check the source to verify that this is the current version before use.

specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 An effective quality assurance system shall be provided to ensure inspection operation performance by personnel who have demonstrated their ability to apply this criteria to the assemblies being inspected, as specified in MIL-A-83377.

4.1.2 Method of sampling for inspection shall be in accordance with MIL-A-83377.

4.2 Tests. Testing shall be in accordance with MIL-A-83377, where applicable.

5. PREPARATION FOR DELIVERY

(Not applicable.)

6. NOTES

6.1 Intended use. This specification is intended to define acceptance criteria of bonded metal faced sandwich assemblies used on aerospace structures.

6.2 Definitions. The defects covered in this specification are defined as follows.

6.2.1 Face-to-core voids and disbonds. Voids are areas in which adhesive is not present. Disbonds are any unbonded areas occurring between the facing and the honeycomb core or between doublers and core.

6.2.2 Metal-to-metal voids or disbonds Any unbonded areas occurring between two solid, nonporous members that are joined by an adhesive bondline.

6 2.3 Porosity. A series of small discontinuities or voids closely spaced. The outer peripheral edges of porosity areas shall be outlined and the porosity areas treated as an unbonded area of similar size and shape.

6.2.4 Core splice void or disbond. Any unbonded area existing in the plane of a bond between two core segments.

6.2.5 Core splice gap. That distance between core segment edges and other core segments or solid members that is filled with core splice adhesive or equivalent.

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:51Z
Check the source to verify that this is the current version before use.

6.2.6 Dents. Local depressions in sandwich faces.

6.2.7 Markoff. A step occurring in the facing of a bonded sandwich.

Custodian:
Air Force - 11

Preparing Activity:
Air Force - 11

Project No. 8040-F092

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:51Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document DoD contractors government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side staple in corner and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE
MIL-A-83376A 25 JAN 79

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes" in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

C16061

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:51Z
Check the source to verify that this is the current version before use.

FOLD

ASD/ENESS
Wright-Patterson AFB, O 45433
OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

U.S. MAIL

ASD/ENESS
Wright-Patterson AFB, Ohio 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:51Z
Check the source to verify that this is the current version before use.